

# CS 660: Data Mining For Decision Making

## Lecture 6 (Week 2)

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Scaling the Heights

# Decision Tree Classifiers: An Alternative Rational Approach

- Often times, we want to use production rules (if – then rules for classification) for making decisions (In decision-making literature they are also referred to as “heuristics”)
- Suppose, we have a set of attributes characterizing a stimulus (E.g., Professor or Student). The idea is to build a decision tree and then use this tree to generate the production rules (if – then rules for classification)
- There are several ways of building decision trees based upon the order in which the attributes are considered
- However, based upon simple mathematics, one could find out the best order of considering attributes in a decision tree and thus figure out the best possible production rules from a training dataset

# Decision Task: Classification as Student or Professor (Training Set)

| NAME of training pattern | Attributes |         |          | Class     |
|--------------------------|------------|---------|----------|-----------|
|                          | HABIT      | EATS    | FOOTWEAR |           |
| T1                       | Gabby      | Baked   | Clogs    | Student   |
| T2                       | Gabby      | Roasted | Sandals  | Professor |
| T3                       | Gabby      | Baked   | Sandals  | Student   |
| T4                       | Quiet      | Fried   | Sandals  | Professor |
| T5                       | Gabby      | Fried   | Clogs    | Student   |
| T6                       | Quiet      | Baked   | Sandals  | Student   |
| T7                       | Gabby      | Fried   | Sandals  | Professor |
| T8                       | Quiet      | Fried   | Clogs    | Student   |

**FIGURE 1.1** Training set for recognizing professors and students. The possible attribute values are 'gabby' and 'quiet' for HABIT; 'baked', 'fried', and 'roasted' for EATS; and 'clogs' and 'sandals' for FOOTWEAR.

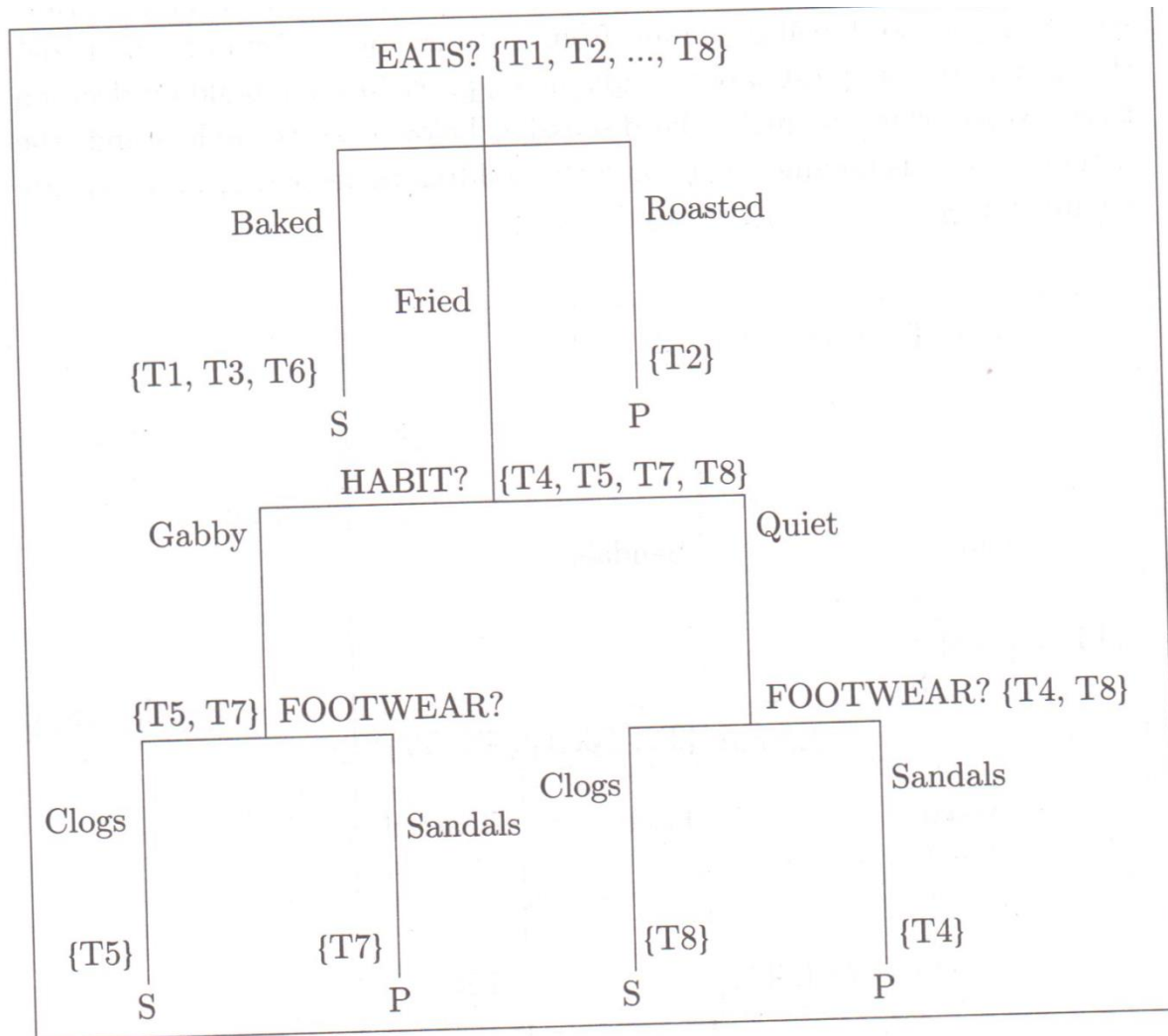
# Decision Task: Classification as Student or Professor (Test/Generalization Set)

| NAME of<br>recall<br>pattern | Attributes |         |          | Class |
|------------------------------|------------|---------|----------|-------|
|                              | HABIT      | EATS    | FOOTWEAR |       |
| R1                           | Quiet      | Baked   | Clogs    |       |
| R2                           | Quiet      | Roasted | Sandals  |       |
| R3                           | Gabby      | Roasted | Clogs    |       |
| R4                           | Quiet      | Roasted | Clogs    |       |

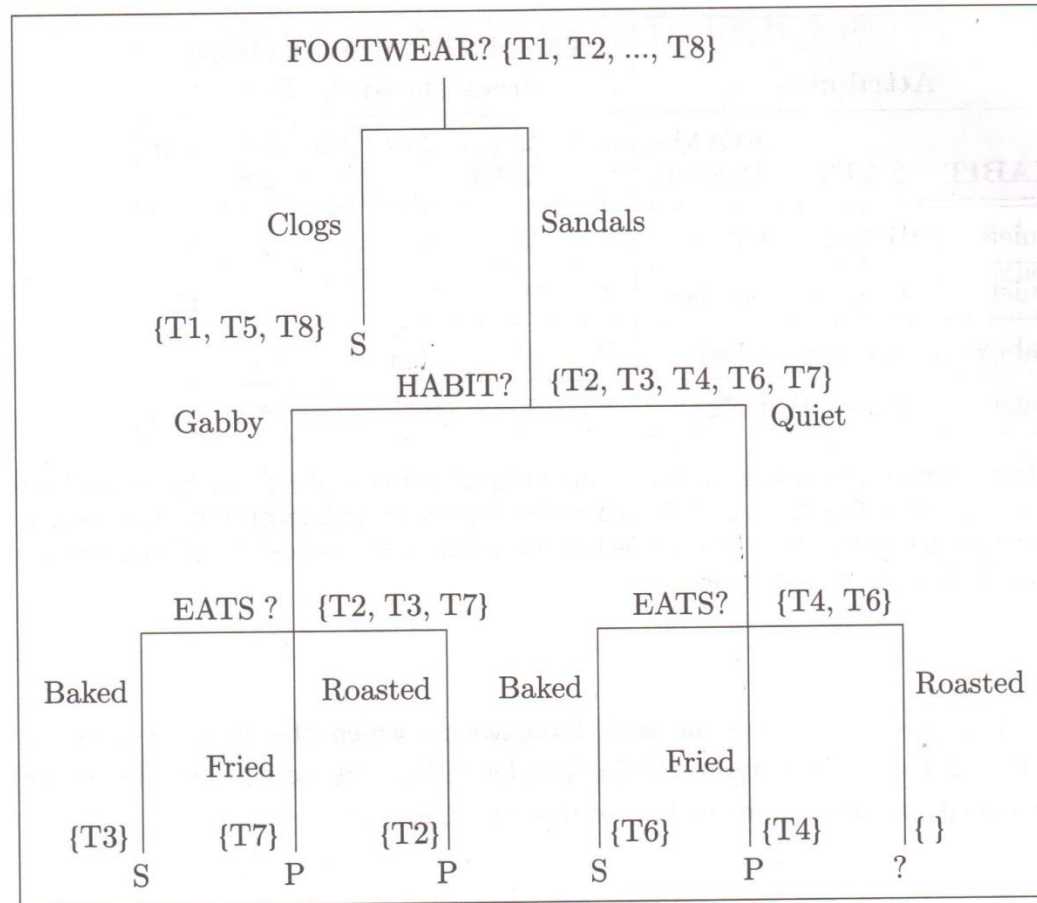
**FIGURE 1.2**

Recall set for recognizing professors and students. The training set is given in Fig. 1.1.

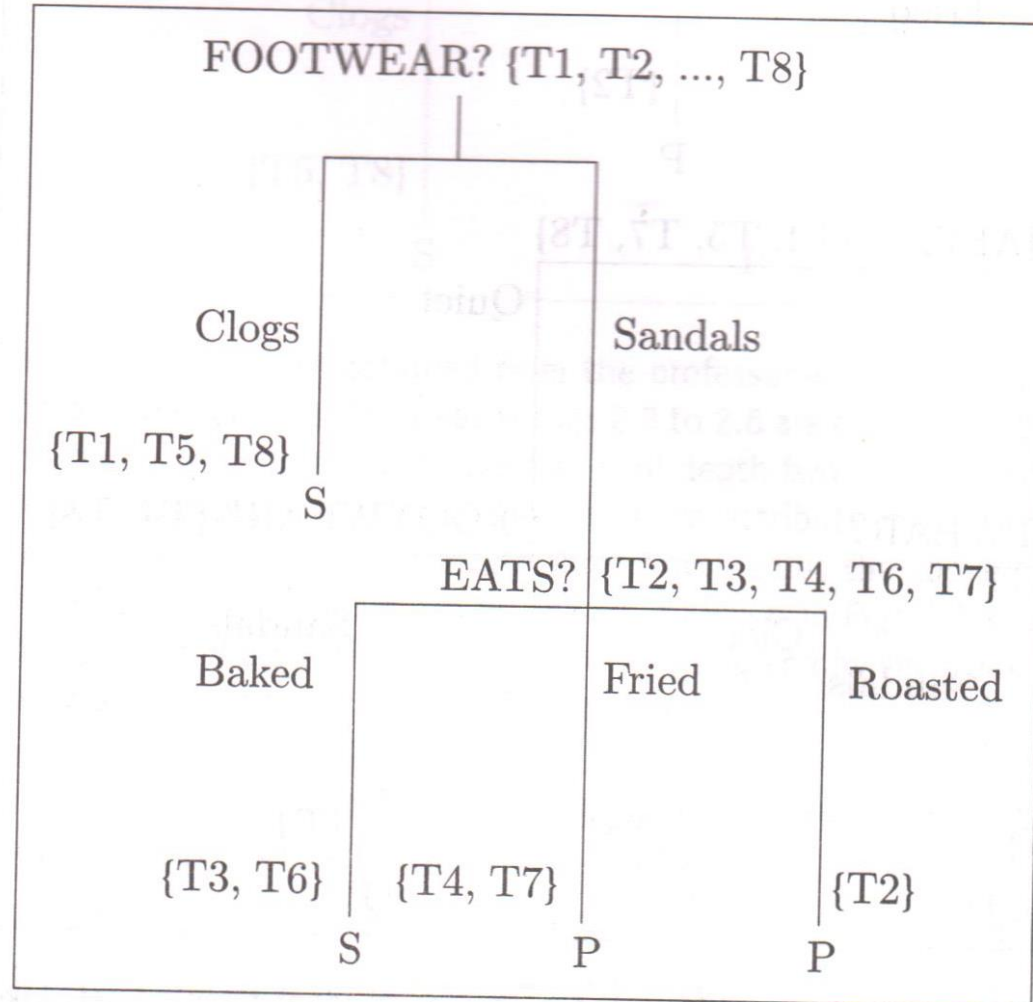
# A Decision Tree



# Another Decision Tree



# Still Another Decision Tree



# How to decide the best sequence of attributes in a decision tree?

Maximize the Ratio of Information Gain =  $(G_A(V_0)) = \frac{I(V_0) - I_A(V_0)}{S_A(V_0)}$

So, the average information needed by attribute  $A$  to split set  $V_0$  into sets  $V_1$  to  $V_n$  is

$$S_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} \quad (2.4)$$

The average information required to classify a pattern in set  $V_0$  into one of the  $m$  classes is then expressed as

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} \quad (2.1)$$

Let  $C_1, C_2, \dots, C_m$  be the  $m \geq 1$  classes to which the training patterns belong. For  $1 \leq k \leq m$  and  $0 \leq i \leq n$ , let  $Y(k, i)$  be the number of training patterns of class  $C_k$  in set  $V_i$ . For  $0 \leq i \leq n$ , let

$$I_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} \quad (2.2)$$

$$Z(i) = \sum_{k=1}^m Y(k, i)$$

# Calculating $I(V_0)$

To evaluate the entropy  $I(V_0)$  of the professor–student training set of Fig. 1.1. The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Three patterns belong to  $C_1$  and five to  $C_2$ . Therefore,

The training set  $V_0 = \{T1, T2, \dots, T8\}$

$Y(1,0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2,0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

From Eqn (2.1) in Section 2.5, the average information required to classify a pattern in  $V_0$  into one of the  $m$  classes is

$$\begin{aligned} I(V_0) &= - \sum_{k=1}^m \frac{Y(k,0)}{Z(0)} \log \frac{Y(k,0)}{Z(0)} \\ &= - \sum_{k=1}^2 \frac{Y(k,0)}{Z(0)} \log \frac{Y(k,0)}{Z(0)} \\ &= -\frac{3}{8} \log \frac{3}{8} - \frac{5}{8} \log \frac{5}{8} \\ &= 0.375 \times 1.415 + 0.625 \times 0.6780 \\ &= 0.5306 + 0.4238 \\ &= 0.9544 \end{aligned}$$

# Splitting on Habit Attribute

To evaluate  $G_{\text{HABIT}}(V_0)$  at the root of a decision tree for the professor-student training set of Fig. 1.1, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since HABIT has two possible values (gabby and quiet), root  $y_0$  has two children  $y_1$  and  $y_2$ . Therefore,  $n = 2$ . To understand the following material easily, one can look at any one of the decision trees shown in Figs 2.3 to 2.6, since each of them examines HABIT at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T2, T3, T5, T7\}$  at node  $y_1$ , where HABIT = gabby

$Y(1, 1) = 2$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 5$  (number of patterns in  $V_1$ )

$V_2 = \{T4, T6, T8\}$  at node  $y_2$ , where HABIT = quiet

$Y(1, 2) = 1$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 3$  (number of patterns in  $V_2$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.9499 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0.0047 \quad [\text{from Eqn (2.5)}]$$

# Splitting on Eats Attribute

To evaluate  $G_{\text{EATS}}(V_0)$  at the root of a decision tree for the professor-student training set, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since EATS has three possible values (baked, fried, and roasted), root  $y_0$  has three children  $y_1$ ,  $y_2$ , and  $y_3$ . Therefore,  $n = 3$ . To understand the following material easily, one can look at either of the decision trees shown in Figs 2.7 or 2.8, since each of them examines EATS at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2, 0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T3, T6\}$  at node  $y_1$ , where EATS = baked

$Y(1, 1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2, 1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 3$  (number of patterns in  $V_1$ )

$V_2 = \{T4, T5, T7, T8\}$  at node  $y_2$ , where EATS = fried

$Y(1, 2) = 2$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2, 2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 4$  (number of patterns in  $V_2$ )

$V_3 = \{T2\}$  at node  $y_3$ , where EATS = roasted

$Y(1, 3) = 1$  (number of patterns in  $V_3$  of class  $C_1$ )

$Y(2, 3) = 0$  (number of patterns in  $V_3$  of class  $C_2$ )

$Z(3) = 1$  (number of patterns in  $V_3$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.5 \text{ [from Eqn (2.2)]}$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.4055 \text{ [from Eqn (2.4)]}$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.3233 \text{ [from Eqn (2.5)]}$$

# Splitting on Footwear Attribute

To evaluate  $G_{\text{FOOTWEAR}}(V_0)$  at the root of a decision tree for the professor-student training set, the following procedure is used. The root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since FOOTWEAR has two possible values (clogs and sandals), root  $y_0$  has two children  $y_1$  and  $y_2$ . Therefore,  $n = 2$ . To understand the following material easily, one can look at either of the decision trees shown in Fig. 2.9 or 2.10, since each of them examines FOOTWEAR at the root.

$V_0 = \{T1, T2, \dots, T8\}$  at node  $y_0$

$Y(1,0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )

$Y(2,0) = 5$  (number of patterns in  $V_0$  of class  $C_2$ )

$Z(0) = 8$  (number of patterns in  $V_0$ )

$V_1 = \{T1, T5, T8\}$  at node  $y_1$ , where FOOTWEAR = clogs

$Y(1,1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )

$Y(2,1) = 3$  (number of patterns in  $V_1$  of class  $C_2$ )

$Z(1) = 3$  (number of patterns in  $V_1$ )

$V_2 = \{T2, T3, T4, T6, T7\}$  at node  $y_2$ , where FOOTWEAR = sandals

$Y(1,2) = 3$  (number of patterns in  $V_2$  of class  $C_1$ )

$Y(2,2) = 2$  (number of patterns in  $V_2$  of class  $C_2$ )

$Z(2) = 5$  (number of patterns in  $V_2$ )

$I(V_0) = 0.9544$  (from Fig. 2.12)

$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k,j)}{Z(j)} \log \frac{Y(k,j)}{Z(j)} = 0.6066 \text{ [from Eqn (2.2)]}$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \text{ [from Eqn (2.4)]}$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{I(V_0) - I_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.3640 \text{ [from Eqn (2.5)]}$$

FOOTWEAR?  $\{T1, T2, \dots, T8\}$

Clogs

Sandals

$\{T1, T5, T8\}$

S

EATS?  $\{T2, T3, T4, T6, T7\}$

Baked

Fried

Roasted

$\{T3, T6\}$

$\{T4, T7\}$

$\{T2\}$

S

P

P

# Splitting on Eats Attribute At Next Level

To evaluate  $G_{\text{EATS}}(V_0)$  at the right child of the root of a decision tree for the professor-student training set, the following procedure is used. FOOTWEAR has already been examined at the root (see Fig. 2.9). The right child of the root is named  $y_0$ . The classes are  $C_1$  (professor) and  $C_2$  (student). Therefore,  $m = 2$ . Since EATS has three possible values (baked, fried, and roasted), node  $y_0$  has three children  $y_1$ ,  $y_2$ , and  $y_3$ . Therefore,  $n = 3$ .

$V_0 = \{T2, T3, T4, T6, T7\}$  at root  $y_0$   
 $Y(1, 0) = 3$  (number of patterns in  $V_0$  of class  $C_1$ )  
 $Y(2, 0) = 2$  (number of patterns in  $V_0$  of class  $C_2$ )  
 $Z(0) = 5$  (number of patterns in  $V_0$ )  
 $V_1 = \{T3, T6\}$  at node  $y_1$ , where EATS = baked  
 $Y(1, 1) = 0$  (number of patterns in  $V_1$  of class  $C_1$ )  
 $Y(2, 1) = 2$  (number of patterns in  $V_1$  of class  $C_2$ )  
 $Z(1) = 2$  (number of patterns in  $V_1$ )  
 $V_2 = \{T4, T7\}$  at node  $y_2$ , where EATS = fried  
 $Y(1, 2) = 2$  (number of patterns in  $V_2$  of class  $C_1$ )  
 $Y(2, 2) = 0$  (number of patterns in  $V_2$  of class  $C_2$ )  
 $Z(2) = 2$  (number of patterns in  $V_2$ )  
 $V_3 = \{T2\}$  at node  $y_3$ , where EATS = roasted  
 $Y(1, 3) = 1$  (number of patterns in  $V_3$  of class  $C_1$ )  
 $Y(2, 3) = 0$  (number of patterns in  $V_3$  of class  $C_2$ )  
 $Z(3) = 1$  (number of patterns in  $V_3$ )

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 0.9710 \quad [\text{from Eqn (2.1)}]$$

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.5218 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.6377 \quad [\text{from Eqn (2.5)}]$$

# Optimal Production Rules

If FOOTWEAR = clogs,  
then the pattern belongs to the class student.

If FOOTWEAR = sandals,  
and EATS = baked,  
then the pattern belongs to the class student.

If FOOTWEAR = sandals,  
and EATS = fried,  
then the pattern belongs to the class professor.

If FOOTWEAR = sandals,  
and EATS = roasted,  
then the pattern belongs to the class professor.



Questions or Comments?